

Complements of Operational Research

1.1 The decision problem

José Firmino Aguilar Madeira

Decision theory: what for?

Decision theory deals with making **rational and consistent** decisions in situations of **uncertainty**.

It provides a set of concepts and techniques to support the decision maker.

Goal

To help choose an action that is **consistent** with the decision maker's alternatives, information, values and logic at the time of the decision.

Examples in engineering and management:

- what capacity to install in a new production line, without knowing the demand;
- which maintenance plan to adopt, without knowing how many breakdowns will occur;
- how much of a seasonal product to order;
- what to invest in (shares, bonds, deposits).

In all of them: we choose **now**; what happens next does not depend only on us.

Good decision $\neq$ good outcome

Question: you are offered a toss of a fair coin. If it lands *heads* you win 30 €; if it lands *tails* you lose 10 €. Do you accept?

Good decision $\neq$ good outcome

Question: you are offered a toss of a fair coin. If it lands *heads* you win 30 €; if it lands *tails* you lose 10 €. Do you accept?

On average you win $\frac{1}{2} \times 30 - \frac{1}{2} \times 10 = 10$ € per toss: accepting is a **good decision**.

Good decision $\neq$ good outcome

Question: you are offered a toss of a fair coin. If it lands *heads* you win 30 €; if it lands *tails* you lose 10 €. Do you accept?

On average you win $\frac{1}{2} \times 30 - \frac{1}{2} \times 10 = 10$ € per toss: accepting is a **good decision**.

It lands tails. You lost 10 €.

Has the decision become bad? **No**: it was the outcome that was bad.

Good decision $\neq$ good outcome

Question: you are offered a toss of a fair coin. If it lands *heads* you win 30 €; if it lands *tails* you lose 10 €. Do you accept?

On average you win $\frac{1}{2} \times 30 - \frac{1}{2} \times 10 = 10$ € per toss: accepting is a **good decision**.

It lands tails. You lost 10 €.

Has the decision become bad? **No**: it was the outcome that was bad.

The distinction that runs through the chapter

A decision is judged by the information available **when it was made**, not by the result that actually happened.

“Good” decisions **do not guarantee** “good” outcomes. Decision theory helps to make good decisions; it does not remove uncertainty.

The elements of a decision problem

Actions $a_1, \dots, a_n$

What the decision maker can choose. The list must be **exhaustive** and the actions **mutually exclusive** (exactly one is chosen).

States of nature $\theta_1, \dots, \theta_m$

Events that may occur and that the decision maker **does not control**. They must be mutually exclusive and describe all situations exhaustively.

Consequences c_{ij}

Each pair (action a_i , state θ_j) is associated with a consequence c_{ij} : its value (gain, cost, utility) for the decision maker.

	θ_1	θ_2	$\dots$	θ_m
a_1	c_{11}	c_{12}	$\dots$	c_{1m}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
a_n	c_{n1}	c_{n2}	$\dots$	c_{nm}

Decision matrix: rows = actions, columns = states.

From the story to the matrix: how many tents to rent?

A company is organising a fair and has to rent tents **before** registrations close. Each tent costs 300 €; each occupied tent brings in 500 €. It is expected that 2, 3 or 4 tents will be needed.

From the story to the matrix: how many tents to rent?

A company is organising a fair and has to rent tents **before** registrations close. Each tent costs 300 €; each occupied tent brings in 500 €. It is expected that 2, 3 or 4 tents will be needed.

- **Actions:** rent $t \in \{2, 3, 4\}$ tents.
- **States:** $d \in \{2, 3, 4\}$ tents are needed.
- **Consequence:** profit
 $L(t, d) = 500 \min(t, d) - 300 t$.

From the story to the matrix: how many tents to rent?

A company is organising a fair and has to rent tents **before** registrations close. Each tent costs 300 €; each occupied tent brings in 500 €. It is expected that 2, 3 or 4 tents will be needed.

- **Actions:** rent $t \in \{2, 3, 4\}$ tents.
- **States:** $d \in \{2, 3, 4\}$ tents are needed.
- **Consequence:** profit

$$L(t, d) = 500 \min(t, d) - 300 t.$$

For example, $t = 4$, $d = 2$: only 2 tents bring in revenue, 4 are paid for: $L = 1000 - 1200 = -200$.

From the story to the matrix: how many tents to rent?

A company is organising a fair and has to rent tents **before** registrations close. Each tent costs 300 €; each occupied tent brings in 500 €. It is expected that 2, 3 or 4 tents will be needed.

- **Actions:** rent $t \in \{2, 3, 4\}$ tents.
- **States:** $d \in \{2, 3, 4\}$ tents are needed.
- **Consequence:** profit

$$L(t, d) = 500 \min(t, d) - 300 t.$$

For example, $t = 4$, $d = 2$: only 2 tents bring in revenue, 4 are paid for: $L = 1000 - 1200 = -200$.

rent t	tents needed d		
	2	3	4
2	400	400	400
3	100	600	600
4	-200	300	800

Renting few guarantees 400 €; renting many may give 800 € or -200 €. **Which to choose depends on the criterion** — this is the topic of 1.2 to 1.4.

What do we know about the states?

Situation	What is known	Where it is studied
Certainty	We know which state will occur: just choose the best consequence in that column.	(trivial)
Uncertainty	The states are known, but nothing about their probabilities.	1.2 Non-probabilistic criteria
Risk	The probability of each state is known (or estimated).	1.3 Probabilistic criteria
Risk with opinion	The probabilities come from the opinion of the decision maker or of an expert.	1.4 Subjective probabilities

Before any criterion: dominance

If $c_{ij} \leq c_{kj}$ for all states j , action a_i is **dominated** by a_k and can be eliminated. In the tents example no action is dominated.

Example 1

A person decides to buy shares in a large technology company. Do they do this because they know what the return on the investment will be?

Example 2

You are challenged to roll a die: you receive 20 € if it shows 1, 2, 3, 4 or 5 and you pay 10 € if it shows 6. Do you roll the die?

Example 3 — investment

An investor can buy shares, bonds or treasury bills. The gain depends on whether the market goes up, stays stable or goes down.

Identify the actions, the states of nature and the consequences, and build the decision matrix.

KEY TAKEAWAYS — 1.1

1. A decision is judged by the information available **when it was made**: good decision $\neq$ good outcome.
2. Elements: **actions** (the decision maker controls them), **states of nature** (does not control them), **consequences** c_{ij} .
3. Actions and states: exhaustive and mutually exclusive lists.
4. The **decision matrix** organises everything: rows = actions, columns = states.
5. **Dominated** actions are eliminated first.
6. The choice criterion depends on what is known about the states: uncertainty (1.2), risk (1.3), opinion (1.4).

Knowing nothing about the probabilities, how do we choose? → 1.2 Non-probabilistic criteria .

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Decision Analysis.
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017 — Decision Analysis.

Resources

- Exercise sheet 1 — Decision theory.